

Software Test Plan (STP)

Project: Chat Application (Socket Programming)

Version: 1.0

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Team Members & Contribution

Name	SRN	Contribution
Pranay Shah	PES2UG24CS366	Section 1 (Introduction), Section 9 (Roles & Responsibilities); overall document structuring, compilation, and final review
Nikhil Mahabala Shekar	PES2UG24CS318	Section 2-4 (Test Items, Features to be Tested, Features Not to be Tested), Section 6 (Test Environment), and functional test cases in Section 13.1
Prarthana Herur	PES2UG24CS367	Section 5 (Test Approach/Strategy including 5.1 Security Validation), Section 10 (Risks & Mitigation), and security/NFR test cases in Section 13.1
Nikhil B Menon	PES2UG24CS317	Section 7-8 (Test Schedule, Test Deliverables), Section 11-12 (Assumptions & Dependencies, Suspension & Resumption Criteria), Section 13.2 (Traceability/RTM), and Section 14 (Test Metrics & Reporting)

Revision History

Version	Date	Author	Change Summary	Approval
1.0	15-09-2026	Pranay Shah, Nikhil Mahabala Shekar, Prarthana Herur, Nikhil B Menon	Initial Test Plan drafted for Chat Application project	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Faculty Guide			

1. Introduction

1.1 Purpose

This document defines the test plan for the Chat Application (Socket Programming) v1.0. It outlines the testing objectives, scope, strategy, resources, schedule, and responsibilities required to verify that the system meets the functional, non-functional, and security requirements defined in the SRS.

1.2 Scope

Testing covers user registration and authentication, real-time one-to-one and group messaging, presence indicators, message history and persistence, file/media sharing, notifications, and administrative moderation. Third-party push-notification provider internals and browser-native WebSocket implementations are excluded from this test plan's scope.

1.3 References

- Chat Application — Software Requirements Specification (SRS) v1.0
- Chat Application — Software Architecture and Design Specification (SAD) v1.0
- OWASP Top 10; RFC 6455 — The WebSocket Protocol; IEEE 829 Standard for Software Test Documentation

1.4 Definitions

- WS — WebSocket; API — Application Programming Interface; JWT — JSON Web Token
- TLS — Transport Layer Security; SRS — Software Requirements Specification
- RTM — Requirements Traceability Matrix; STP — Software Test Plan; UAT — User Acceptance Testing

2. Test Items

- User Registration & Authentication module
- Real-Time Direct Messaging module
- Group Chat / Chat Rooms module
- Presence & Typing Indicators module
- Message History & Persistence module
- File & Media Sharing module
- Notifications module
- Administrative / Moderation interface

3. Features to be Tested

Features mapped to SRS requirement IDs:

Req ID	Feature / Requirement
CHAT-F-002	Authenticate user and issue JWT session token
CHAT-F-011	Real-time one-to-one direct message delivery
CHAT-F-022	Broadcast message to all connected room members
CHAT-F-030	Online/offline presence indicator
CHAT-F-041	Retrieve paginated message history
CHAT-F-051	Reject oversized/disallowed file uploads
CHAT-NF-001	Message delivery latency $\leq 1s$ (95th percentile)
CHAT-NF-002	Support ≥ 500 concurrent WebSocket connections per instance
CHAT-SR-001	TLS enforced for all WebSocket and REST connections
CHAT-SR-004	Sanitisation of message content to prevent XSS

4. Features Not to be Tested

- Internal implementation of third-party push-notification providers (e.g. Firebase Cloud Messaging delivery infrastructure)
- Browser-native/OS-native WebSocket protocol implementation
- Underlying cloud provider infrastructure provisioning and network hardware
- Third-party object storage service internals used for media storage

5. Test Approach / Strategy

Levels

- Unit tests (module/service level, e.g. Auth Service, Messaging Service)
- Integration tests (WebSocket Gateway ↔ Messaging/Presence/Room services, service ↔ database/Redis)
- System tests (end-to-end chat flows across client and server)
- Acceptance tests (UAT with representative end-user scenarios)

Types

- Functional testing (core chat features)
- Regression testing (after each build/change)
- Performance testing (message latency, concurrent connection load)
- Security testing (authentication, authorization, input validation)
- Usability testing (UI clarity, accessibility, responsiveness)

Entry Criteria

A stable build is delivered; test environment (server, database, Redis) is provisioned and reachable; test data (seed accounts/rooms) is available.

Exit Criteria

100% of planned test cases executed; zero critical or high-severity defects open; all acceptance criteria in the SRS satisfied; NFR and security targets met or formally waived.

5.1 Security Validation

- Validate JWT handling — correct rejection of expired or tampered tokens on both REST and WebSocket calls
- Verify TLS 1.2+ is enforced; confirm plaintext (ws:// / http://) connections are refused
- Verify passwords are stored using salted, adaptive hashing (bcrypt/argon2) and never in plaintext
- Fuzz-test message content and file-upload fields for injection and malformed-input handling
- Verify server-side sanitisation of message content prevents stored/reflected XSS
- Rate-limiting / denial-of-service testing on login and message-send endpoints
- Penetration testing of authentication and authorization flows, including room-moderation privilege checks

6. Test Environment

Hardware: Standard developer workstations/laptops for server and client testing; representative mobile devices for responsive UI testing.

Software: Chat backend (Node.js + Socket.IO/ws or Python Flask-SocketIO), PostgreSQL, Redis, and modern browsers (Chrome, Firefox, Edge, Safari).

Tools: Selenium/Playwright (UI automation), Postman (REST and WebSocket API testing), JMeter/Artillery (performance and concurrent-connection load testing), Jira (defect tracking).

Test Data: Dummy user accounts, seeded chat rooms and group memberships, and pre-populated message history.

7. Test Schedule

Milestones:

- Test case design: 18-Sep-2026
- Environment setup: 20-Sep-2026
- Test execution start: 22-Sep-2026
- Test execution end: 26-Sep-2026
- UAT: 27-Sep-2026 to 28-Sep-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Prarthana Herur	Prepare test plan, coordinate execution, own security validation sign-off
Test Engineer	Nikhil Mahabala Shekar	Design & execute functional and integration test cases, log defects
Test Engineer	Nikhil B Menon	Design & execute performance/NFR test cases, maintain traceability matrix, log defects
Developer	All team members	Support defect fixes, triage, and re-testing
Product Owner	Pranay Shah	Approve test results, sign off on release readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke-test builds from the development team
Test environment downtime (server/DB/Redis)	Maintain a backup environment on a cloud VM
Dependency on third-party push-notification provider	Engage provider sandbox early; maintain test stubs/mocks for notification calls
Flaky WebSocket connections during load testing	Use dedicated, isolated load-test environment with monitored network conditions

11. Assumptions & Dependencies

- The chat backend build (WebSocket Gateway, Messaging/Presence/Room services) will be stable and available for the scheduled test window
- Test data (seed accounts, rooms, message history) will be provided or seeded before execution begins
- Redis and PostgreSQL test instances will be provisioned and reachable throughout testing

12. Suspension & Resumption Criteria

Suspend testing if: the test environment is unavailable for more than 4 hours, or the build is unstable enough to block more than 30% of planned test cases.

Resume testing if: blocking defects are resolved and the environment is confirmed stable.

13. Test Case Management & Traceability

13.1 Test Cases

A minimum set of 13 test cases is defined below, covering functional requirements (authentication, messaging, rooms, presence, history, media), non-functional requirements (latency, concurrency), and security requirements (TLS enforcement, XSS sanitisation).

TC ID	Req ID	Description	Preconditions	Steps	Expected Result	Priority
TC-Auth-01	CHAT-F-001	Register with a unique username/email and valid password	Registration endpoint reachable	Submit registration form with valid, unique details	Account is created and success response returned	High
TC-Auth-02	CHAT-F-002	Login with valid credentials	Registered account exists (TC-Auth-01 passed)	Submit login with correct email/password	Valid JWT token is returned; login succeeds	High
TC-Auth-03	CHAT-F-003	Account lockout after repeated failed logins	Registered account exists	Submit 6 consecutive incorrect password attempts	6th attempt returns an account-lockout message	High
TC-Msg-02	CHAT-F-011	Real-time direct message delivery	Two authenticated users online with open WS connections	User A sends a direct message to User B	User B receives the message within 1 second	High
TC-Msg-03	CHAT-F-012	Offline message queuing and delivery	Recipient is offline; sender is online	Sender sends a message while recipient is offline, then recipient reconnects	Recipient receives all queued messages, in order, after reconnecting	High
TC-Room-03	CHAT-F-022	Broadcast message to room members	A room with 3+ online members exists	One member sends a message to the room	All other online members receive the broadcast message	High
TC-Pres-01	CHAT-F-030	Online/offline presence indicator updates	Two users are contacts/room members	User A disconnects and reconnects	User B's UI reflects the status change within 3 seconds	Medium
TC-Hist-02	CHAT-F-041	Paginated message history retrieval	A conversation with 50+ historical messages exists	Request message history with pagination parameters	Correct paginated set is returned in chronological order	Medium
TC-Media-02	CHAT-F-051	Reject oversized file upload	User is authenticated in an active conversation	Attempt to upload a file exceeding the configured size limit	Upload is rejected with a clear error message	Medium
TC-Perf-01	CHAT-NF-001	Message delivery latency under normal load	Load-testing environment configured	Send a sustained stream of messages under normal simulated load	95th percentile end-to-end latency is $\leq$ 1 second	High
TC-Perf-02	CHAT-NF-002	Concurrent WebSocket connection capacity	Load-testing tool (JMeter/Artillery) configured	Ramp up to 500 concurrent WebSocket connections on one server instance	All 500 connections remain stable with latency target intact	High
TC-Sec-01	CHAT-SR-001	TLS enforcement on all connections	Test client capable of attempting plaintext connections	Attempt to connect via ws:// and http:// instead of wss:// / https://	Plaintext connection attempts are refused by the server	High
TC-Sec-04	CHAT-SR-004	XSS sanitisation of message content	Two authenticated users in a conversation	Send a message containing a <script> tag payload	Message renders as inert text on the recipient's client, not executable script	High

13.2 Requirements Traceability Matrix (RTM)

The RTM ensures every SRS requirement maps to at least one executable test case.

Req ID	Requirement	Test Case(s)
CHAT-F-002	Authenticate user	TC-Auth-02, TC-Auth-03
CHAT-F-011	Real-time direct message delivery	TC-Msg-02, TC-Msg-03
CHAT-F-022	Broadcast message to room	TC-Room-03
CHAT-F-030	Online/offline presence	TC-Pres-01

Req ID	Requirement	Test Case(s)
CHAT-F-041	Retrieve paginated message history	TC-Hist-02
CHAT-F-051	Reject oversized/disallowed uploads	TC-Media-02
CHAT-NF-001	Message delivery latency $\leq 1s$ (p95)	TC-Perf-01
CHAT-NF-002	≥ 500 concurrent WebSocket connections	TC-Perf-02
CHAT-SR-001	TLS enforced on all connections	TC-Sec-01
CHAT-SR-004	XSS sanitisation of message content	TC-Sec-04

14. Test Metrics & Reporting

Metrics collected: % test cases executed, % passed/failed, defect density, defect aging, requirement coverage.

Reports: Daily execution status report; final Test Summary Report at the end of the execution cycle.

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		